conviction_tilt_legacy

equity strategy | family: equity-neutralized | status: reference | derived from `conviction_tilt`

What this strategy does

Skip the middle 15% either side of average, lean toward the strongest surprises, balance by size -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. Concentrate capital where the surprise is largest; neutralize size so it is not a hidden bet.

Design notes (from the strategy's source file).

Conviction Tilt -- skip the unsure middle, lean on the strongest signals, balance by size.

Built after diagnosing why rank-weighting had the better Sharpe but the worse return: it traded ~4x more positions and saturated the liquidity/leverage caps. This strategy drops the low-conviction middle of the SUE-rank distribution (trim 0.15), leans weight toward the extremes (tilt), keeps size neutrality, and when the leverage cap binds drops the LOWEST-conviction positions first instead of shaving everything equally.

parameter	value
base_unit_fraction	0.002
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

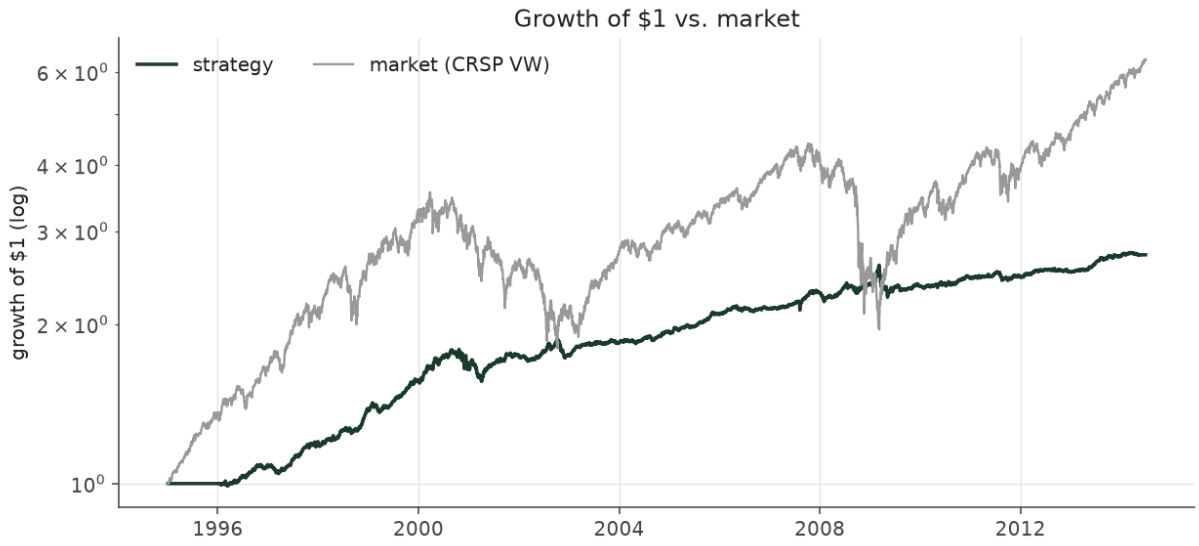
metric	conviction tilt legacy	market (CRSP VW)
Annualized return	5.25%	9.94%
Annualized volatility	4.41%	19.35%
Sharpe (NAV-pct)	1.18	0.59
Sortino	1.70	
Max drawdown	-13.1%	-55.5%
Calmar	0.40	
Final NAV ($10M start)	$27,077,292	
Worst day	-2.43%	
% positive days	52.3%	
Sharpe 1996-2001	1.74	
Sharpe 2002-2007	1.31	
Sharpe 2008-2013	0.60	

Correlation of daily returns with the market: **0.05**.

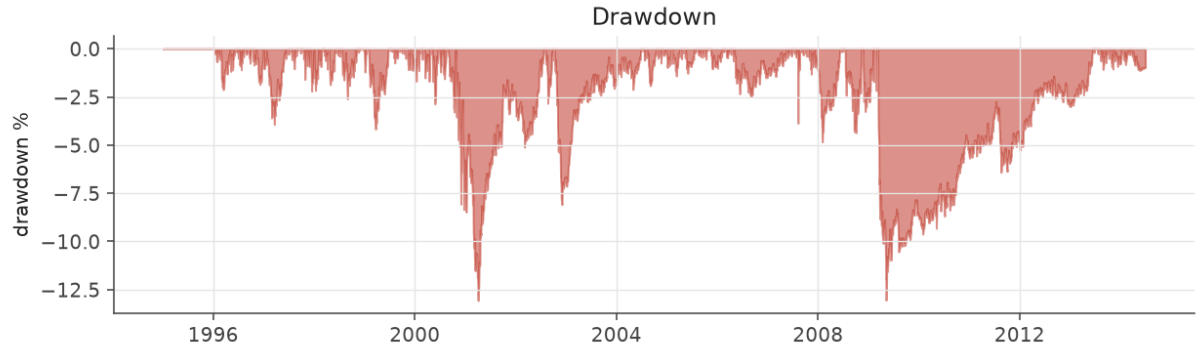
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.18

• best return among equity: `market_boost_max` (0.116); this one: 0.0525

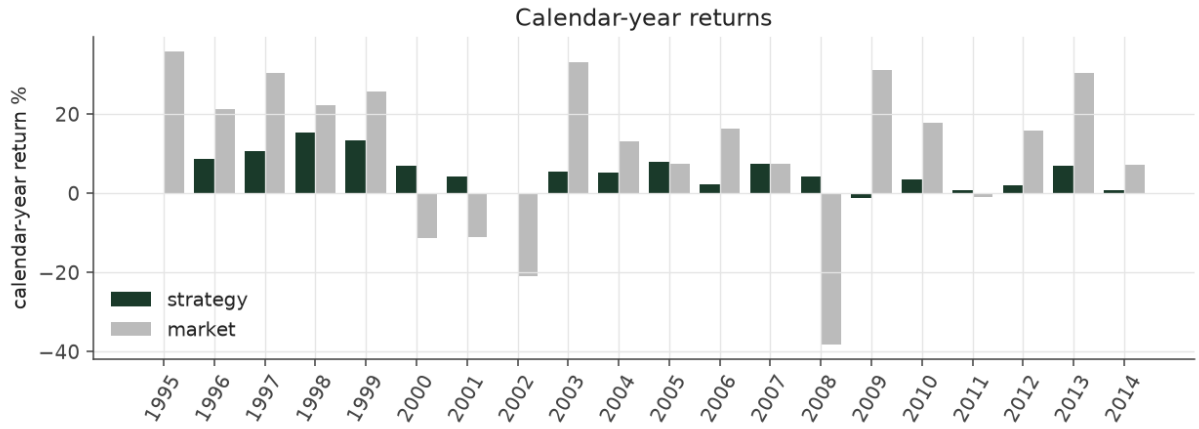
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.402



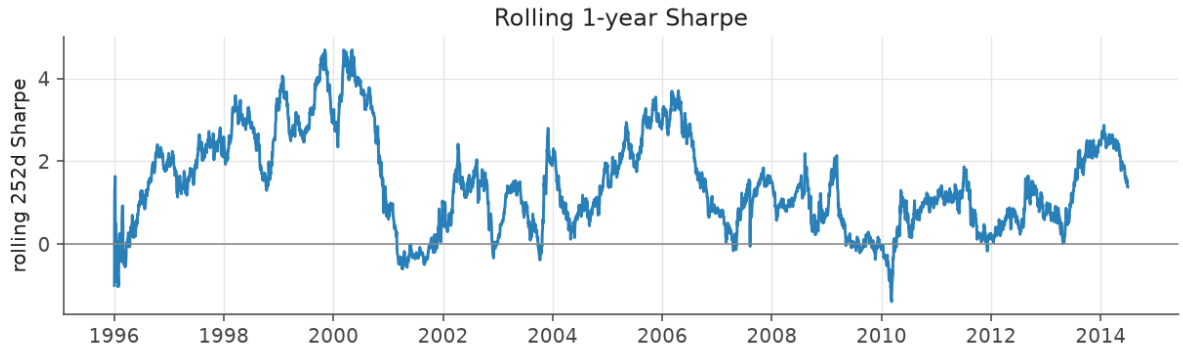
NAV (log scale)



Drawdown from running peak

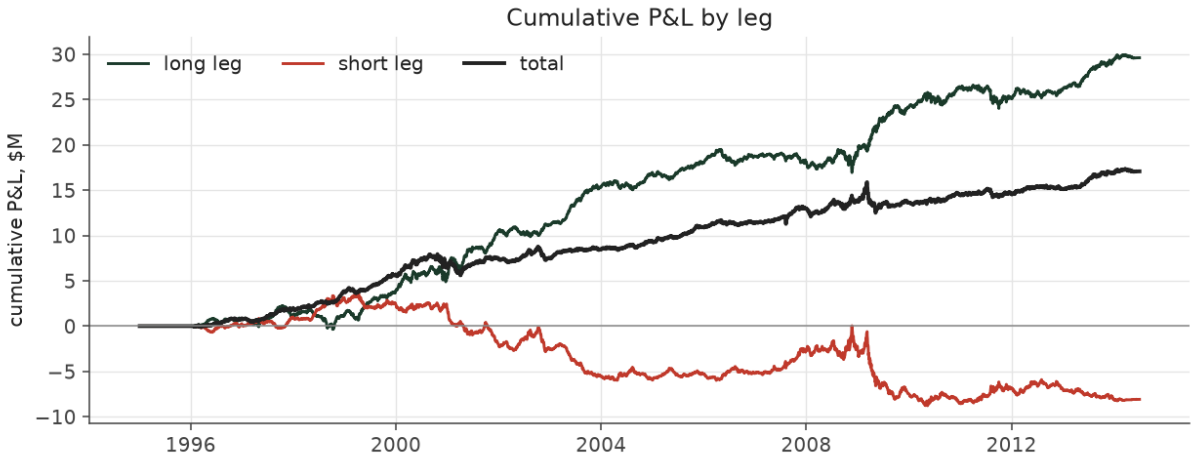


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	29.61	-8.08
P&L after costs ($M)	27.48	-10.41
of which common (average-event) component ($M)	15.95	-16.41
of which excess / PEAD component ($M)	13.10	8.76
excess minus costs ($M)	10.98	6.45
gross return per $ traded (bp)	297.54	-77.04
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.21	-0.29

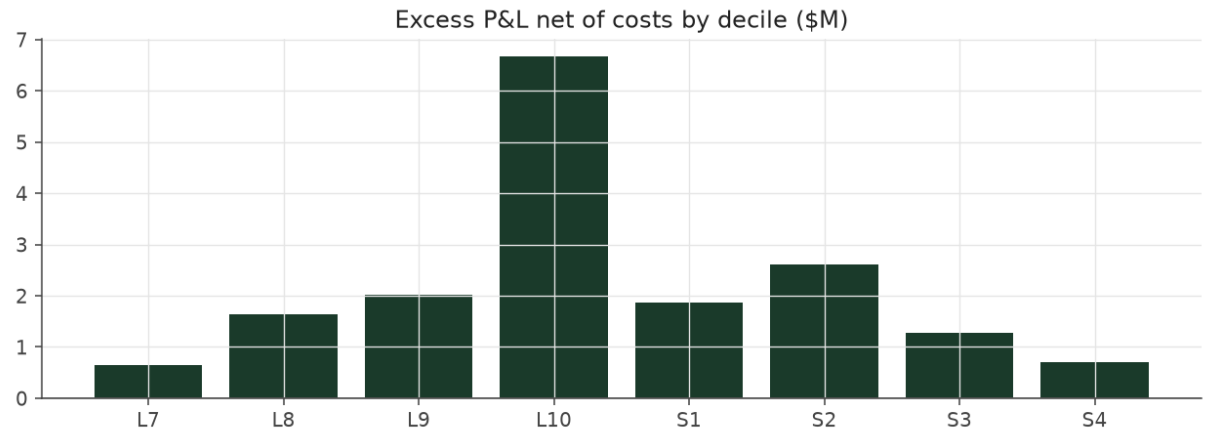
Daily P&L correlation between the two legs: **-0.78**.

• **The short leg loses money in absolute terms** (-8.1M before costs) while the long leg earns +29.6M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 16.0M / 16.4M. Stripping that common component, the surprise signal itself earned +13.1M on the long side and +8.8M on the short side.

• The two legs' daily P&L are -0.78 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.21 long / -0.29 short vs 1.18 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	5,109	54.41	2.00	0.77	0.13	0.64	3.7%	117.71
long	8	15,516	217.64	5.62	2.12	0.48	1.64	9.4%	75.55
long	9	17,587	322.33	6.68	2.68	0.67	2.01	11.5%	62.42
long	10	17,044	400.78	13.19	7.52	0.84	6.67	38.3%	166.49
short	1	16,800	412.85	-4.55	2.81	0.95	1.86	10.7%	45.08
short	2	17,012	330.93	-2.84	3.33	0.73	2.61	15.0%	78.82
short	3	15,637	230.86	-2.49	1.77	0.50	1.27	7.3%	54.96
short	4	7,653	74.51	-0.52	0.84	0.15	0.69	4.0%	93.31



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,635	277.38	4.88	6.56	1.07	5.49	31.5%	197.86
2	22,249	437.74	5.40	6.60	1.26	5.33	30.6%	121.88
3	24,545	445.39	3.82	4.55	0.93	3.62	20.8%	81.28
4	26,161	444.26	1.90	2.56	0.69	1.87	10.7%	42.16
5	26,768	439.55	1.08	1.57	0.48	1.09	6.2%	24.71

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	11,936	204.63	4.83	6.17	1.02	5.15	29.5%	251.49
2	22,441	467.78	4.88	6.82	1.40	5.41	31.1%	115.73
3	24,862	468.24	3.94	5.10	0.94	4.16	23.9%	88.86
4	26,282	458.63	2.03	2.50	0.64	1.86	10.7%	40.48

5	26,832	444.91	1.39	1.27	0.45	0.83	4.7%	18.59

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	42,328	715.87	-0.58	3.89	1.32	2.56	14.7%	35.83
41-60d	24,752	462.93	2.26	2.69	1.02	1.67	9.6%	35.96
61-90d	33,610	629.62	5.66	6.24	1.44	4.80	27.5%	76.17
>90d	11,668	235.90	9.74	9.04	0.67	8.38	48.1%	355.12

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,693	480.63	7.06	8.09	1.01	7.08	40.6%	147.33
2002-07	37,768	691.10	5.43	7.05	1.48	5.57	32.0%	80.66
2008-13	41,897	872.59	4.59	6.71	1.96	4.75	27.2%	54.41

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,073	387.69	3.73	2.99	0.85	2.15	12.3%	55.34
Chems	2,426	41.78	0.15	0.29	0.08	0.21	1.2%	50.53
Durbl	2,657	48.63	0.55	0.80	0.11	0.69	4.0%	142.22
Enrgy	5,779	101.46	1.40	1.75	0.20	1.55	8.9%	152.99
Hlth	10,938	200.75	2.46	2.37	0.48	1.88	10.8%	93.75
Manuf	11,395	202.91	1.11	1.91	0.43	1.48	8.5%	72.85
Money	16,527	302.05	2.08	3.91	0.64	3.26	18.7%	108.03
NoDur	4,812	85.02	0.71	0.82	0.17	0.65	3.7%	76.39
Other	17,162	317.77	2.40	3.41	0.75	2.67	15.3%	83.88
Shops	11,409	209.54	2.37	2.77	0.45	2.31	13.3%	110.40
Telcm	3,788	69.14	0.10	0.49	0.14	0.35	2.0%	51.03
Utils	4,392	77.58	0.01	0.33	0.14	0.20	1.1%	25.51

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	37,309	677.56	3.73	5.21	1.33	3.88	22.3%	57.29
low vol	37,771	683.62	6.69	8.70	1.64	7.06	40.5%	103.23
mid vol	35,726	647.23	6.25	7.66	1.41	6.25	35.9%	96.62

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	5.25%	0.00%
0.5%	1.1	5.03%	0.22%
1.0%	2.2	4.80%	0.45%
2.0%	4.3	4.31%	0.94%
5.0%	10.9	2.51%	2.73%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.